Response to reviewers

Introspection Dynamics with Mutation in Additive Games

Harry Foster, Vince Knight, Sebastian Krapohl

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Dear Professor Zaccour,

We thank both reviewers for their careful reading of the manuscript and for their constructive comments. We respond to each comment below; reviewer text is set in italics, and our responses follow in upright type. Where a comment has led to a change in the manuscript we describe the change.

Reviewer 1

Main comments

> *In the abstract, perhaps the introspection dynamics process could be better explained to be more accessible to readers who are not familiar with it.*

We have rewritten the opening of the abstract to describe the process explicitly: at each time step a randomly selected player compares their current payoff to the payoff an alternative action would have given, and switches with a probability increasing in the difference. Following a minor comment below, we have also removed all formulas from the abstract.

> *I think the manuscript would sound stronger with a slight framing reordering: the authors could first say they extend introspection dynamics to include mutations, and from there they rederive results for additive games.*

Done. The abstract and the introduction now lead with the extension: we first state that we extend introspection dynamics to include per-player, per-action mutation and player-specific selection intensities, and then present the additive-game results of Couto and Pal as rederived, and generalised, from the extended dynamics.

> *The Introduction could include some more references. Think of a reader who hasn't carefully read References [1] and [2]: it would be hard for them to know how the present model relates to previous ones, such as, for example, other EGT models for asymmetric games or logit-response dynamics.*

Done. The introduction now relates introspection dynamics to the logit-response dynamics of evolutionary game theory in economics (Blume, 1993; Alós-Ferrer and Netzer, 2010), notes evolutionary approaches to asymmetric games (McAvoy and Hauert, 2015), and adds the exploration dynamics of Traulsen et al. (2009) for the role of mutation as exploration, error, and behavioural noise.

Theorem 1's proof is intuitive (and perhaps even an obvious result). However, it remains somewhat verbal. Perhaps the authors could complement it with a more formal argument. For additive games, because players' updates are independent of what others do, we can define a smaller state space for each player, {C, D}. [...] It follows trivially that the stationary distribution of such a transition matrix is {p_i, 1-p_i}.

Done, following exactly the suggested argument. Section 3 now defines, for each player, the selection kernel K_i on their own action set: the stochastic matrix governing a selected player's next action, well defined precisely when the game is additive. Theorem 1 verifies directly that the product of the per-player stationary distributions is stationary for the full chain, and that positive mutation makes the chain irreducible and aperiodic, so the stationary distribution is unique. For two actions the two rows of K_i coincide at $(p_i,\, 1-p_i)$, which is the reviewer's argument; the closed form for the marginal (Theorem 2) and the explicit product measure (Theorem 3) follow. This per-player-chain framework also supports the alternative mutation scheme raised by Reviewer 2.

Even though one cannot obtain closed formulas for non-additive games, it would still be nice to see the effect of the mutations for those games too. Would you observe similar patterns for non-linear PGGs? (E.g., can mutation increase cooperation frequency when cooperation is not expected?)

Done, in a new section ('Non-additive games') with a new six-panel figure. We take the non-linear public goods game with synergy or discounting parameter w (Hauert et al., 2006) as a case study and compute exact quantities from the full 2^N linear system, for parameters at which defection dominates the linear game. The section reports the following. First, the $w = 1$ values coincide with our closed form, a numerical check of the additive theory. Second, with discounting the additive pattern persists qualitatively: mutation raises low cooperation towards $\frac{1}{2}$, though no longer affinely. Third, and answering the reviewer's question directly, with synergy mutation produces high cooperation where none is expected: the game is bistable, the mutation-free strong-selection dynamics started from defection never cooperate, yet any positive mutation rate makes the chain ergodic and strong selection then drives cooperation up towards $1-\mu$ rather than down to μ. Mutation acts as an equilibrium-selection device in non-additive games, and we note the parallel with the role of rare mutations in stochastic evolutionary dynamics. Fourth, beyond additivity there can be an optimal, strictly positive mutation rate: for a bistable synergy game the exact p_C is non-monotone in μ, peaking at $\mu^* \approx 0.02$, and the optimum moves and its peak falls as selection strengthens; our mutation-selection balance result rules this out for any additive game, where p_C is affine in μ. Finally, we show that additivity is exactly the point at which the players' actions decouple: heatmaps of the pairwise correlations, for the same heterogeneous group as our validation figure (contributions $\alpha_i = i$), show that every pairwise correlation is identically zero at $w = 1$ despite the heterogeneity, weakly negative under discounting, and positive and increasing in both players' contributions when the two strict equilibria compete. Beyond additivity a player's long-run behaviour is tied to their co-players' parameters, in contrast to the additive case.

Reference [2] provides a closed formula for the stationary distribution for additive games with any number of strategies. Would your extension with mutations work similarly in the general case? Perhaps it would be worth including such a result if possible (even in the simplest case of symmetric mutations).

Done, and for arbitrary player- and action-specific mutation probabilities rather than only the symmetric case. We have gone further and restructured the paper so that the general result is now the lead result: the setup of Section 2 is stated for finite action sets A_i, and Section 3 first establishes that for any additive game in which player i, when selected, mutates to action b with probability $\mu_{ib} > 0$, the stationary distribution of the full chain is the product of the per-player stationary distributions (Theorem 1); the two-action closed form follows as a specialisation (Theorems 2 and 3). Computing the stationary distribution therefore reduces from one linear system of dimension $\prod_i L_i$ to N systems of dimension L_i. We also record what is special to two actions: only there do the rows of the per-player transition matrix coincide, so for three or more actions the marginals are given by N small linear systems rather than by a closed form.

Minor comments

> *The manuscript is generally well-written, but it would benefit from another proofreading, as there are a few typos and missing words/commas.*

We have made a proofreading pass alongside the corrections below, fixing in particular a comma splice in the discussion of M_1 and M_2, one in the proof of Theorem 1, and inconsistent equation referencing (all display equations are now referenced with `\eqref`).

> *Introspection dynamics is generally defined for games with any number of strategies. Therefore [...] it should read "Introspection dynamics, in which each player compares their payoff to their payoff under an alternative action," [...] And specify that the linear system that needs to be solved to obtain the stationary distribution has dimension 2^N for the particular case with 2 strategies.*

Done. The abstract and the introduction now say 'an alternative action', and the introduction states that the linear system has dimension 2^N, more generally L^N when each player has L actions.

> *Abstract: I would personally avoid using formulas in the abstract, or else all parameters should be defined (e.g., it's not clear what μ is in the abstract).*

We have removed all formulas from the abstract. The mutation probabilities are now described in words (payoff-independent, player-specific probabilities of switching to cooperation or to defection), and the closed form for the public goods game is described as having N terms rather than 2^N.

> *Page 2: I'd write 'general Prisoner's Dilemma' instead of 'classical (RPST) Prisoner's Dilemma'.*

Done, in both the introduction and Section 2.

> *Indentations, e.g., after Eq 1.*

Done. Text that continues a sentence after a display equation (after equations (1) and (3)) is no longer indented.

> *Page 3: It may not be clear what $(1, 1)$ means in $\Delta f_1((1, 1))$ because that notation was not defined before.*

Done. The sentence now reads "writing (a_1, a_2) for the state in which player 1 plays a_1 and player 2 plays a_2" before the notation is used.

> *Page 3: "Additivity for player 1 requires the equality $T - R = P - S$". Additivity actually requires this equality for all players. And perhaps it's best to show the matrix form for the general Prisoner's Dilemma too. The acronym 'PD' has not been defined.*

Done. The text now reads "Additivity requires, for each player, the equality", the general Prisoner's Dilemma is displayed in matrix form as G_3, and the acronym PD is defined at first use. (The example matrices, previously M_1, M_2, M_3, are now G_1, G_2, G_3, to avoid a collision with the total mutation probability M_i.)

> *Page 3: "(both differentials equal c)." To be precise, it's c_1 and c_2, unless you want to refer to the symmetric case.*

Done. The text now reads "for player i, both differentials equal c_i".

> *Page 3: "To define introspection dynamics we introduce the following two quantities:" I wouldn't call the Fermi function and the mutation scheme 'quantities'. Also, using μ_0 for the probability of mutating to cooperation and μ_1 for the probability of mutating to defection is a counterintuitive notation given you denote cooperation as $a = 1$ and defection as $a = 0$ before.*

We now write 'components' rather than 'quantities'. The mutation probabilities have also been renamed throughout: they are now indexed by the target action, μ_{iC} for mutation towards cooperation and μ_{iD} for mutation towards defection, consistent with the general notation μ_{ib} for $b \in A_i$; the notation μ_{i0}, μ_{i1} no longer appears.

> *Page 4: At times, you refer to C as a state. [...] To be precise, avoid calling C a state. (An alternative would be defining another possible state space for each player that is simply {0,1} or {C, D}.)*

Done. The wording now refers to selections 'resulting in action C' rather than placing a player 'in state C', and, following the reviewer's suggested alternative, each player's action set now serves as a formal per-player state space: Section 3 defines the selection kernel K_i on A_i (which is $\{C, D\}$ in the two-action case) as part of the restructured proofs (see our response to the main comments above).

> *Page 5: "without mutation the chain degenerates to a point mass at the dominant-action profile". Consider complementing this observation with an example to make it clearer for all readers.*

Done. We now give the donation game M_1 as an example: each $\delta_i = c_i > 0$, so as $\beta \to \infty$ the mutation-free stationary distribution collapses to a point mass on the all-defect profile.

> *Page 5: "Mutation therefore extends the exact analysis to arbitrarily strong selection." This sentence might also not be fully clear for readers not familiar with introspection or Markov chains [...]*

Done. We have expanded this paragraph along the lines the reviewer sketches: without mutation the chain loses ergodicity in the strong-selection limit, so the limit must be taken after computing the stationary distribution at finite selection intensity, as in [1]; mutation keeps the switching probabilities bounded away from 0 and 1, so the chain stays ergodic at any selection intensity and a unique stationary distribution always exists.

> *Page 8: In the second paragraph, you refer to the "stationary cooperation level of the linear public goods game in Proposition 3 of [2]." Note that the equation given in Proposition 3 of [2] is not exactly for the stationary cooperation level, but rather the general stationary level for any state. Also, this paragraph is just one sentence – consider joining it with the previous one.*

Done. The sentence now states that the product measure of Theorem 2 reduces to the stationary distribution given, for every state, in Proposition 3 of [2], with the cooperation level following; it has been joined to the preceding paragraph.

> *Figure 1: In the legend, you mean Corollary 2 instead of 1. In the caption, it should read "All values were obtained using [3]."*

Done. The figure has been regenerated with the corrected legend, and the caption now reads "All values were obtained using [3]."

> *Table 1: "Both columns to four decimal places".*

Done. The caption now reads "Both columns are given to four decimal places."

> *Page 9: It can be confusing to use the words 'benefactor' and 'beneficiary'. Make it clearer what you mean by each.*

Done. We now define the terms once, immediately after Lemma 1: a player is a *net beneficiary* when $r_i > N$, since each unit they contribute returns more than a unit to them from the pool, and a *net contributor* when $r_i < N$. These two terms are used consistently throughout; 'benefactor' and 'non-beneficiary' no longer appear.

> *Remark 3: I believe it's not exactly a "negative quantity ϕ_i", but a negative sign stemming from ϕ_i', to be more precise.*

Done. The remark now reads "a negative sign stemming from the derivative $\phi_i' < 0$".

> *Figure 2: The order in which each panel is mentioned in the text is switched. The right panel is perhaps unnecessary because it's not showing any variation (and that can be easily seen through the equations). Why not a plot that varies over the mutation rates instead, for example?*

Done, on both counts. The right panel has been replaced by a plot of p_i as a function of the symmetric mutation rate $\mu \in [0, \frac{1}{2}]$ for $r_i < N$, $r_i = N$, and $r_i > N$, illustrating the affine mutation-selection balance of Proposition 2, with all lines meeting at $p_i = \frac{1}{2}$ when $\mu = \frac{1}{2}$. The surrounding text has been reordered so that the panels are discussed left, centre, right.

> *Page 10: I personally prefer when a Conclusion section doesn't require the reader to know all the notation introduced in the body text.*

Done. The conclusion has been rewritten so that, together with the introduction, it is self-contained: the results are stated in words (the players behave independently in the long run, each player's choice is forgetful, and the computation reduces to one small linear system per player) rather than through the notation of the body text.

Reviewer 2

> *I'm worried that their result (Theorem 1) holds for a particular kind of mutation model where players can mutate to both the actions (A or B) independently, no matter which action they are playing currently. [...] It turns out that if you only allow mutation to the other strategy, the statement $p(A \to B) = p(A \to A) = p_A$ (Theorem 1) is no longer true even if the two mutational probabilities are the same. This means that Theorem 2 (the factorization result) is not immediately true for this other (equally reasonable*

or arguably more reasonable) mutational scheme. That said, Theorem 2 could hold regardless; I did not check that. Could the authors justify the necessity of self-mutation (biologically or game theoretically)? I suggest that the authors perform some analysis of this alternate mutational scheme also.

The reviewer is correct that under mutation restricted to the other action the forgetful property fails: the probability that a selected player's next action is C then differs, by the mutation rate, between the two current actions. We have added a subsection ('An alternative mutation scheme') analysing this scheme in full, with action-dependent rates ν_{iC}, ν_{iD}. The factorisation that the reviewer suspected 'could hold regardless' does hold: our general product-measure theorem uses only that a selected player's next action depends on their own current action, so it applies verbatim, and the marginals follow from the two-state balance,

$$p_i = \frac{(1-\nu_{iC})\phi_i(\delta_i) + \nu_{iC}}{1 + \nu_{iC}\big(1-\phi_i(\delta_i)\big) + \nu_{iD}\,\phi_i(\delta_i)}.$$

Moreover, we prove that for symmetric rates the two schemes are equivalent: the alternative scheme with rate ν_i has exactly the same stationary distribution as our scheme with $\mu_{iC} = \mu_{iD} = \nu_i/(1+\nu_i)$. Whether a mutating player may re-select their current action is therefore a reparameterisation of the mutation rate rather than a substantive modelling choice, which we believe addresses the concern that self-mutation is 'a convenient trick that makes the maths easy' in the strongest possible way. For asymmetric rates the schemes genuinely differ, and the subsection records the difference. We have also added references justifying both conventions: redrawing from the full action set is the convention of exploration dynamics (Traulsen et al., 2009) and of the payoff-independent mutations used in stochastic equilibrium selection (Kandori et al., 1993), whilst mutation to the other action only is the bit-flip mutation of genetic algorithms (Holland, 1992; Goldberg, 1989). The figure isolating the role of mutation now also overlays the alternative scheme at the reparameterised rate, illustrating the coincidence of the two schemes.

Regarding the overall contribution: since the reviewer's report the manuscript has also gained the general result for arbitrary finite action sets (Theorem 1), formal restructured proofs, and a numerical section on non-additive games (equilibrium selection by mutation, an optimal mutation rate, and the decoupling of the players exactly at additivity), which we hope addresses the comment that the paper was short in its previous form.

Thank you again for the constructive reports.

Harry Foster, Vince Knight, and Sebastian Krapohl